HDOC DAY – 45

ALGORITHM:

Main Algorithm

Step 1: Start.

Step 2: Display the menu:

- Change date format
- Print all substrings
- Exit

Step 3: Read the user's choice.

Step 4: Use a switch statement to perform the selected operation.

Step 5: If choice is 1, call changeDateFormat().

Step 6: If choice is 2, call printSubstrings().

Step 7: If choice is 3, terminate the program.

Step 8: For any other choice, display "Invalid choice".

Step 9: Repeat the menu until the user selects Exit.

Step 10: Stop.

Algorithm 1: Change Date Format

Function: changeDateFormat()

Step 1: Start the function.

Step 2: Read the date in dd/mm/yyyy format.

Step 3: Extract the day, month, and year using sscanf().

Step 4: Store the month names in an array:

Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec

Step 5: Check whether the entered month is between 1 and 12.

Step 6: Obtain the last two digits of the year using year % 100.

Step 7: Print the date in dd-mmm-yy format.

Step 8: Return to the main function.

Algorithm 2: Print All Substrings

Function: printSubstrings()

Step 1: Start the function.

Step 2: Read the input string.

Step 3: Find the length of the string using strlen().

Step 4: Use an outer loop to select the starting position of each substring.

Step 5: Use a second loop to select the ending position.

Step 6: Use a third loop to print all characters from the starting position to the ending position.

Step 7: Repeat until all possible substrings are printed.

Step 8: Return to the main function.

C PROGRAM:

```
#include <stdio.h>
#include <string.h>

// Function prototypes
void changeDateFormat();
void printSubstrings();

int main()
{
   int choice;
```

```
do
{
   printf("\n----- MENU -----\n");
   printf("1. Change date format\n");
   printf("2. Print all substrings of a string\n");
   printf("3. Exit\n");

   printf("Enter your choice: ");
   scanf("%d", &choice);
   getchar();

   switch (choice)
   {
      case 1:
         changeDateFormat();
         break;

      case 2:
         printSubstrings();
         break;

      case 3:
         printf("Exiting program...\n");
         break;
```

```
            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}


// Function to change date format
void changeDateFormat()
{
    char date[20];
    int day, month, year;

    char *months[] = {
        "Jan", "Feb", "Mar", "Apr",
        "May", "Jun", "Jul", "Aug",
        "Sep", "Oct", "Nov", "Dec"
    };

    printf("Enter date in dd/mm/yyyy format: ");
    fgets(date, sizeof(date), stdin);
```

```
    if (sscanf(date, "%d/%d/%d", &day, &month, &year) != 3)
    {
        printf("Invalid date format!\n");
        return;
    }

    if (month < 1 || month > 12)
    {
        printf("Invalid month!\n");
        return;
    }

    printf("Date in new format: %02d-%s-%02d\n",
           day, months[month - 1], year % 100);
}


// Function to print all substrings
void printSubstrings()
{
    char str[100];
    int i, j, k;
    int length;
```

```
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    // Remove newline added by fgets
    str[strcspn(str, "\n")] = '\0';

    length = strlen(str);

    printf("Substrings are:\n");

    for (i = 0; i < length; i++)
    {
        for (j = i; j < length; j++)
        {
            for (k = i; k <= j; k++)
            {
                printf("%c", str[k]);
            }
            printf("\n");
        }
    }
}
```

TEST CASE TABLE:

Test Case	Choice	Input	Expected Output	Result

1	1	12/10/2026	12-Oct-26	Pass
2	1	01/01/2025	01-Jan-25	Pass
3	1	25/12/2026	25-Dec-26	Pass
4	1	15/13/2026	Invalid month!	Pass
5	2	abc	a, ab, abc, b, bc, c	Pass
6	2	AB	A, AB, B	Pass
7	2	xyz	x, xy, xyz, y, yz, z	Pass
8	3	—	Exiting program...	Pass
9	5	—	Invalid choice! Please try again.	Pass